

An Astrologer's Day

---- **R.K Narayan**

R.K. Narayan, the winner of Sahitya Academy Award, is one of the most famous Indian writers in Indian English writing. He achieved international acclaim as a prolific and perfect storyteller. He wrote several stories which fascinated the readers throughout the world. An Astrologer's Day is one of them. This is the first beautiful short story in Narayan's anthology called *Malgudi Days*.

An Astrologer's Day is a very short story. It is very typical in its theme and style. It deals with the themes of fear, revenge, freedom, deception, guilt, relief and the ironies of life. It combines suspense, realism, and thriller genres. Here the central character, the Astrologer, faces some kind of crisis and resolves it at the end. The interesting plot, the element of suspense, the logical climax, the figurative language and the underlying meaning behind the story are the important elements of this story. These elements have been artistically interwoven in the story and that is why this story becomes impressive and interesting.

The narrator of the story seems to be omniscient. It seems that he knows everything and describes the things in third person. He describes a day in the life of an astrologer. The astrologer lays out all of his professional equipment, which consisted of a dozen cowries' shells, a square piece of cloth, a notebook and a bundle of Palmyra writing. He uses sacred ash and vermilion on his forehead as makeup and a turban on his head to make himself appear more mystical. By chance he becomes an astrologer. Thus he is a stranger to the stars as much his innocent clients. A perfect picture of an astrologer has been presented and his shrewdness is very well described by Narayan. The astrologer's place of work finds fantastic expression in this story.

The narrator narrates the past story of the astrologer. Circumstances compel him to leave his small village silently without any previous plan. One day when the astrologer starts to pack up at the end of the day, an unusual client appears in the scene to consult the astrologer. The story takes a new turn when he compels the astrologer to answer his specific questions. The astrologer succeeds in bargaining and the interesting part of the story begins. The stranger is a smoker who uses the match stick to light his cheroot. The astrologer catches a glimpse of his face in that light of the match stick as it was darkness there. What the astrologer says hereafter is the subject of amazement.

The astrologer then starts telling the past story of the stranger. He tells the stranger that once he was stabbed by one of his friends. That friend then pushed him into a well to die. It was the truth of the life of the stranger. The astrologer then calls his name - Guru Nayak. Guru Nayak is completely confused. The stranger asks about the location of his enemy. The astrologer advises that he should give up his hunt because the attacker is now no more in the world. Four months ago in a far off town he has been crushed under a lorry. Nayak is pleased by this news. The astrologer also advises Guru Nayak to go home and stay up there and never travel southward again. If he does so, he will surely be killed.

The story takes another twist when the astrologer reaches home and after a nice dinner he tells his wife his past story. He says that when he was a youngster, got drunk, gambled and got into a quarrel. He adds that in a fit of rage he stabbed Nayak and pushed him into the well and left him to die. Due to this he left his village, settled here, and married her. And this was how he could so correctly talk of Guru Nayak's troubled past. All these years he had thought that the blood of a man was on his hands. But now the man he thought he had killed is alive. Thus he can sleep with a light heart now. The story thus ends.

Unit. II

BUILDING A NEW STATE

Dr. A.P.J. Abdul Kalam

Dr. A.P.J. Abdul Kalam was born at Rameshwaram in Tamil Nadu. He was the eleventh president to India and was conferred India's highest civilian award 'Bharat Ratna' in 1997. During his tenure from 2002-2007 he was popularly known as "people's president".

In the present lesson, "Building a new state" (taken from 'Ignited Minds') we will come to know his experiences and findings during the four visits to Jharkhand state. Dr. Kalam had already visited Jharkhand state for three times before the month of September 2001. The first two visits brought him very close to the core competence that the state possessed. He had been made Patron of the Science and Technology Council of the state and his purpose of that fourth visit was to work out a developmental programme in the areas of herbs, forest products and other natural resources in order to increasing income to the state. According to Dr. Kalam all the components for developing Jharkhand state such as – fertile area, good rainfall, tall trees, rich vegetation and industrious people were there. Kalam expressed in his own words that they made some progress on the fourth visit. Later, along with his members of the mission he started

for Bokaro, the steel city despite the rough weather. While flying in a helicopter he was struck by looking the clean environment by totally forgetting himself.

In his addressing to the Chinmaya Vidyalaya students Dr. Kalam reinforced his keen observation by expressing a statement that the state had all the needed wealth. He tried to motivate all the young children to work very hard in the “mission mode” by making best use of minerals in abundance in their state. Dr. Kalam thought their contribution to the state itself could be tremendous. He also felt that the education system needed to be improved very urgently when he went round the exhibition put up by the same school children.

Dr. Kalam continued his other engagements after the function at Chinmaya Vidyalaya. He had a meeting due at the town hall where he had to speak on “Jharkhand’s Core Competence and Industries.” During the meeting a very pertinent question was posed by the audience about raw material that was exported from many ports to other countries. In answer, Dr. A.P.J. narrated a conversation that he had with Dr. Jose Paul, chairman of the Mormugao Port Trust during his short stay in Goa. Dr. Kalam came to know from the words of Dr. Jose Paul that 30 million tones of iron ore was exported from only Mormugao Port alone that to for a low price. The same ore if it is utilized here, it would generate far more income because of “value addition” which is not properly focused and blindly ignored by India.

The next day Dr. Kalam travelled to the Bokaro steel plant and was impressed with the scale of the plant and hundreds of hardworking men over there but he was disappointed a lot to see that there were no industrial estates around it, utilizing the steel produced there to make various products. He was told that the setting up of industrial estate came under state purview. He was not happy for such compartmentalized thinking and for the fragmented governance.

Soon after this, while travelling by the flight back to Delhi, Dr. Abdul Kalam kept on thinking how Jharkhand state could best be helped. According to him what were needed were a few major missions to transform the state and a time-frame. In addition to this, the state and the centre would need to make an integrated effort. He firmly hoped that the “mission mode” operations would bring rapid development at minimum cost. Throughout the lesson we can sense that Dr. Kalam and his mission members were highly dedicated and totally committed to their assigned task.

Unit. III

Water: The Elixir of life – Essay

--- C.V. Raman

Sir C.V. Raman is an Indian physicist. He was awarded the Nobel Prize in 1930 for his work on molecular scattering of light and for the discovery of the Raman Effect. In this present lesson Raman vividly made an effort to bring about the importance of water. Since the time immemorial, man has sought in vain for an imaginary elixir of life, the divine Amrita. A drop of amrita was thought to confer immortality. But in the words of Sir C.V. Raman, the true elixir of life the plain water.

Sir C.V Raman recollects the day he was standing on the line which separates the Libyan Desert from the valley of the Nile in Egypt. On one side was visible a sea of billowing sand without any life and on the other side lay one of the greatest, most fertile and densely populated areas to be found anywhere on the earth. It was teeming with life and vegetation. It was due to the water of the river Nile. The ancient civilization of Egypt was created and sustained by the life giving waters which comedown year after year with unfailing regularity. He gives this example to emphasize that the common water is the most potent and the most wonderful thing on the face of our earth.

There is nothing which aids so much to the beauty of the country side as water. It may be just a little stream trickling over the rocks or it may be a little pond by the way side where the cattle quench their thirst of an evening. The rain fed tanks that are so common and play a vital role in south Indian agriculture. It is a cheering sight to see the sun rise or sunset over one of them. Water in a landscape may be compared to the eyes of a human face. It reflects the mood of an hour, being bright and gay when the sun shines. It turns to dark and gloomy when the sky is overcast. Therefore, C V Raman compares water in a landscape to the eyes in a human face.

One of the most remarkable facts about water is its power to carry silt in suspension. This is the origin of the characteristic colour of the water in rain fed tanks. The soil formed by silt thus deposited is very fertile. The flow of water can also play a destructive part and wash away the soil which is the foundation of all agriculture. If it is allowed to proceed unchecked, it will show the most disastrous effects on the life of our country. The problem of soil erosion is one of the serious problems in various countries and especially in many parts of India. Soil erosion occurs in successive steps. It gradually washes away the earth, making agriculture impossible. Sudden bursts of excessively heavy rain resulting in a large run of

surplus water are the principal factors in causing soil erosion. It can be checked by terracing of the land, construction of bunds to check the flow of water, the practice of contour cultivation and the planting of appropriate types of vegetation.

Water is the basis of all life. Every animal and plant contains water in its body. Water is essential for the body, moisture in the soil is equally important for the life and growth of plants and trees. The conservation and utilization of water is fundamental for human welfare. Much of Indian agriculture depends on monsoon. It is clear that the adoption of techniques preventing soil erosion would help to conserve and keep the water where it is wanted.

In India, where we depend on seasonal rainfall, an immense quantity of rain water runs off the ground. The collection of rain water and utilizing them is very important. Vast areas of land which at present are scrub jungle. By courageous and well-planned action, it could be turned into fertile and prosperous country. The planning of trees in every possible way and the development of forest is one of the most urgent needs of India. Such plantation would check soil erosion.

In one sense, water is the commonest of liquids and on the other hand contains all amazing properties which are responsible for its unique power of maintaining plant and animal life. The investigation of the nature and properties of water is therefore should be of the highest scientific field of interest.

4. The wood rose

■ AbburiChayadevi

AbburiChayadevi was a Telugu fiction writer; she won the Sahitya Academy Award in Telugu for the Year 2005, for her work Tana Margam. Chaya Devi was the wife of the late AbburiVaradarajeswara Rao, himself a writer, critic and former Chairman, Official Languages Commission. She was born AbburiChayadevi on 13 October 1933 in Rajahmundry, East Godavari district, Andhra Pradesh and died on 28 June 2019 in Hyderabad, Telangana.

The Wood rose' is a story of an aged widow & her relationship with her son & daughter- in law. She feels disconnected with them as there is no warmth in their relationship. She's on perpetual 'rest' as the daughter- in law doesn't let her in the kitchen. The narrator sits in the evening everyday at balcony and watches passersby.

She thought that the old age would be a state of happiness, short sleep, reclining in an armchair, playing with her grandchildren, narrating tales and prattling with her grandchildren, gossiping with friends and relatives. She realised that there is a great difference between spending old age in urban and rural areas. Wood rose is creeper, her daughter-in-law said that wood rose is like bell shaped yellow flowers in the middle of the flowers there is a sandal wood colour and finally it is like roses. These flowers do not give fragrance but they are very beautiful. The narrator thought that instead of planting wood rose if we plant Sannajaji creeper planted, one can adorn one's hair, and can be given to newly married couples and if ash guard plant is planted the fruits can be distributed to neighbours. The narrator suggested to her daughter in law to grow brinjal or lady's finger plants instead of crotons and cacti plants.

One day her son Gopalam pull down the wood rose creeper as it spread over the grill, veranda, begins to block the light and crawled densely over the terrace. The narrator shouted at his son Gopalam to stop cutting but he did not heed the words of his mother and shouted at her.

She remembered the love and affection of Gopalam that he had showed on the narrator when he was young. He could not bear his father words on her mother. He used to say that he would spare her from drudgery of cooking. Finally she thought that would be pushed a side like the stone grinder to a corner. She thought that affection between human being should not grow as densely as that wood rose creeper; if it does their children would uproot it.

PROGRESS

---- St. John Greer Ervine

Progress" by St. John Greer Ervine is a successful specimen of a one act play. It has not only a unity of theme (the abolition of war) the unities of time; place and action are also meticulously maintained. Thematically, "Progress" is written against the background of the First World War in which thousands were butchered and many more became disabled for life. It left many mothers childless and incredible number of widows and orphans. This insensible devastation has created diverse reactions in social thinking, and "Progress" by the Irish playwright St. John Greer Ervine is just the product of one of the strongest of these sentiments, namely the anti-war feeling of the post war modern age.

Through the basic conflict between two ideas- the point of view of the war mongering arm dealers (represented through the character of Prof. Corrie) and that of the peace loving universal mothers of our house (represented by Mrs. Meldon) the drama has also sufficiently exposed the motive behind wars –which is to perpetuate and extend the class interest the of the war –mongering imperialists and exploiters, who sit safe and secure –far behind the fighting lines, when the poor sacrifice their lives in the trenches. The conflict is finally resolved in favour of the latter point of view through the murder of Prof. Corrie. Corrie-peace here affirms itself through a little violence; true progress, it seems, necessitates the destruction of the means of destruction. Indeed, there is a tragic severity in the revelation of Mrs.Meldon’s austere and unfathomed bereavement, which is at the same time heart –rending and awe –inspiring.

Again, we have only three characters, and though the different stages of structural development are skillfully maintained, the action-free from anything extraneous or redundant-rushes headlong to the crisis from the very stage of the exposition. There is a surprise turn of events in the denouement, and Prof. Corrie’s murder by his sister, though tragic, is highly dramatic. At the outset, we find Professor Corrie is talking excitedly about his destructive bomb and is busy with his experiment. He is happy and proud as his experiment has proved a success. His sister Mrs. Meldon is grief – stricken as she has lost her only son Eddie in the First World War. She lost her husband also in the war. Corrie advises her not to think about the past. Corrie rejoices over his invention as it can destroy a vast city in a few seconds. He says that his invention will make war in future over in a few hours. He is also proud that the success of the war will depend on who strikes first and what kind of weapons he uses. Mrs. Meldon gently reminds him that his invention will lead to the death of hundreds of young men like her son Eddie. But he ignores her words and gloats over his invention and the fame and wealth. He says that he will sell his bomb to the government which grants him the highest amount of money. When Mrs. Meldon is convinced that her brother will not suppress his invention at any cost, she decides to do away with him along with his invention. She destroys the sheets of paper on which Corrie has written the formulae. Read More Drama But Corrie is not upset. He says that the formulae remain embedded in his brain and he can reproduce them easily. Mrs. Meldon cannot tolerate his inhumanity any more. When he stoops to pick up the sheets, she stabs him on the back with a knife and kills him. She feels that this is justifiable revenge against the murderer of her son on the occasion of his death anniversary.

Though essentially a one-act play, Ervine's "progress" thus leaves a lasting effect of a full length drama. Since the play has little physical action, the interest of the play is maintained by its sharp dialogue, conveying the ideological conflict of the characters. Each character has his or own individual manner and style of speech, keeping parity with the character's bent of mind and point of view. Read More Drama The playwright, who mainly wrote for the Abbey theatre Dublin amply shows his good knowledge of stage technique. The drama may also have an autobiographical element, as the dramatist too had to go to war and face its horrors and atrocities. In fact he was severely wounded in France and eventually lost his leg.

NON-DETAILED

G.D. NAIDU

G.D Naidu is commonly referred as the 'Edison of India' for his significant contribution in the field of technology, industry and development. Though he was some what mischievous at school and home, over time he came to appreciate the importance and value of proper and comprehensive education. A defining moment of his life came when he was still a youngster. He was inspired to watch a foreigner who had ridden a motorcycle through his native village Kalangal. He wanted to ride and understand its mechanics. His new found passion and eventual desperation forced him to leave for Coimbatore.

Coimbatore was a city bustling with industry and education. It was a city ripe with opportunity. Naidu with his burning passion wanted to earn money and became a waiter at a hotel. He worked diligently and saved money. Meanwhile he came to know the whereabouts of the foreigner who happened to be a British surveyor. He met him and placed all his money at his feet and explained about his history and passion for motorcycle. The surveyor convinced by Naidu's determination, gave him his motor cycle.

Naidu was ecstatic and enjoyed riding his motor cycle in his free time. He understood the mechanics of it by taking apart it piece by piece of it and refitting it. In this process he became very familiar with the engine and procured a job as a mechanic. Now he understood how things operated and wanted to use this knowledge to invent new things. In 1920 he expanded his horizons and set up his own transport business. He also dabbled in cotton business and made some money. He purchased an automobile coach and took to riding between Pollachi and Pallani. Over time his transport business, Universal Motor Transport expanded and it owned the most efficient fleet of six hundred transport vehicles.

Naidu began travelling and his passion for technology extended to cameras and films and in 1935 he filmed the funeral of King George in London. He even secured manufacturing contracts from Adolf Hitler. This helped him to develop and manufacture razors. Though he was unsuccessful at politics, he enjoyed a period of comfort and luxury and got a Rolls Royce car as a gift.

In his mid forties Naidu collaborated with D. Balasundaram to create India's first indigenous motor. National Electric Works, a sister concern of UMS group, made the motor in 1937. He continues to invent many things like a tamper-proof vote-recording machine, a kerosene run fan and a razor on the German model. He developed a distance adjuster for film cameras. His keen interest in other fields of research like agriculture resulted in identifying new varieties of cotton, maize and papaya. His dedication and passion for invention was well appreciated by great people like Sir C.V. Raman and M. Visweswaraya.

After his retirement from his automobile company Naidu set up many charitable and philanthropic foundations that helped the poor and encouraged the youth. He started the first polytechnic college in Coimbatore which in due course became Government College of Technology. Sir C.V Raman paid a glowing tribute at Naidu's funeral and considered him to be a great educator, entrepreneur in many fields of engineering and industry and a warm hearted being. Such people are rare indeed.

G. R. GOPINATH

G. R. Gopinath is an entrepreneur and founder of Air Deccan, India's first low cost air line. He is a retired Captain of the Indian army, an author, a politician and a multimillionaire. Though he was born in a village in Karnataka, he rose to a high position because of the encouragement of his parents, support of his teachers and more over by dint of his hard work and pro-active attitude.

Gopinath attended a Kannada medium school till fifth standard. The school headmaster encouraged all the students of the fifth class to appear for Sainik School Entrance Examination conducted by the Ministry of Defence. Unfortunately the examination was in English medium and all the students of Kannada medium failed as they did not know English. The headmaster wrote stern letter to the Ministry of Defence in Delhi and forced them to conduct re-examination in Kannada. Gopinath took the exam and passed with flying colours. He learnt the importance of proactive attitude and hard work to steer things one's way. He studied in Sainik School in Bijapur and later joined National Defence Academy in Khadakvasla, Pune. When he was commissioned as a captain in the Indian army he was stationed in several places in the north India. He spent eight long

years in the army and even served in the Bangladesh Liberation War in 1971. During this period he visited his relatives in U.S and this travel added grit and confidence he needed later.

Gopinath left army after eight years of active service and purchased ten acres of land in his village. His determination paid off and his farm prospered. In 1996 he received the Rolex Laureate Award for his sericulture farm. He began to branch out and show his preference to various other things. He secured a deal with Royal Enfield and started Malnad Mobikes business and also opened a hotel in Hasan. As he was a man of numerous talents, he started writing articles on agriculture and this led him starting up two new ventures- Water Resource Management Consultancy and Supplying Irrigation Systems.

Financially secure and with a good and solid reputation to back him, Gopinath decided to join politics. He sold off all his business and concentrated on the Jain Irrigation Dealership- Espak Agro. He joined BJP Party and became the president of the party in Hasan. After his electoral disaster he backed out of politics. After his exchange visit to China, he met his good old friend in the army, Colonel. A. J. Sam who had taken freelance job as a helicopter pilot G. Gopinath realized the value of air transport. He, A J Sam and Colonel Jayanth Poovaiah started Deccan Aviation company. Gopinath excelled during this time and as a result of their sincere efforts in 2003 Air Deccan, the low cost air line, was born.

From a single helicopter and a firm belief and stern management, Gopinath had gone from a farmer to a successful businessman. His air business expanded under the banner LCC (Low Cost Carriers), developed in to major player in the air industry. In 2006 he was knighted and got the highest civilian award the French Government. In 2007, Vijay Malya, sensing the great potential of Air Deccan made a deal with Gopinath to merge Air Deccan with his Kingfisher Airlines. He founded Deccan Charters and Deccan 360, which focus on cargo and the express logistics business. He also plans on founding another airline with many foreign players. The future lies wide open for proactive people like G.R. Gopinadh.

SUDHA MURTHY

Sudha Murthy, wife of Infosys Narayana Murthy, is a woman of many talents. She is a gold medalist in Computer Science from IIS. She is a prolific writer, a good administrator, philanthropist, a perfect loving wife and embodiment of Indian womanhood. Sudha Murthy, daughter of a physician from Karnataka, completed her M. E. in Computers from Indian Institute of Science and stood first and received a gold medal. She became the first female Engineer hired by TELCO. When her application was rejected on grounds of sex, she made a complaint to company

Chairman J.R.D. TATA. She was granted a special interview and was selected. She met N. R. Narayana Murthy at TELCO in Pune. Later Narayana Murthy moved to Patni Computers. He and Sudha got married in 1978. She made many sacrifices while her husband Murthy setup Infosys along with five others. She extended all the needed financial help and with a seed money of Rs.10000/- Infosys got started and later she involved in its development.

She took up a job with Walchand Industry to support the growth of Infosys. She was involved in the development of Infosys Technologies Ltd. in various capacities. She worked for some time as the Head, Department of Computer Science in a college and a visiting Professor at the PG Centre of Bangalore University. In 1996 she co-started Infosys Foundation and was the Trustee of Infosys Foundation and later she became the Chair person. She was a member of public health care initiatives of the Gate Foundation. She has founded several orphanages, participated in rural development efforts like providing Kannada Government students with computer and Library facilities. She established 'The Murthy Classical Library of India, at Harvard University. In memory of her maternal grandparents two institutions of higher learning, were endowed and inaugurated by Infosys foundation

In 2004 Sudha Murthy got the Rajyalakshmi award for her outstanding social worker. In 2006 she was awarded the Padma Shri for her varied contributions and R. K. Narayan Award for Literature in 2011. She, along with Justice Santosh Hedge, was conferred honorary LLD for her contribution to promoting formal legal education and scholarship in India. Sudha Murthy is a prolific fiction writer and through her fictional narratives she propagated her views on charity, hospitality and self realization. Her 'How I Taught My **Grand-mother** to Read and Other stories' has been translated into fifteen languages. 'The Day I Stopped Drinking Milk', 'Wise or Otherwise', and 'Old Man and his God' are some of her notable books. As story teller at heart, she shuts herself away from everything else and deeply involves in her project.

She is a woman of compromises and adjustment and tries to bridge the gap of generations and comes close to the youngsters like his son-in-law and daughter-in-law. She believes that marriage works well if you don't have many expectations from each other. She has a great passion for travel and different vegetarian cruises. Though she is modern in everything, she is traditional in housekeeping. She admits that she could give wings to her philanthropy only because of her husband Murthy's support.

Homi Jehangir Bhabha

Homi Jeehangir Bhabha was born to a wealthy Parsi family in 1909. Bhabhas had a very close connection with Tatas. Homi was brought up in an atmosphere of academics and learning. He did his schooling at the Cathedral and John Common School in Bombay and in 1924 he sat for the senior Cambridge examination and took admission to the Elphinstone College in Bombay. Later he joined The Royal Institute of Science and in 1927 he got admission into Gonville and Caius College Cambridge. His father and his uncle wanted to see him an engineer at the Tata iron and steel plant at Jamshedpur.

Bhabha was mesmerized by the intellectual environment that ran in Cambridge and drawn to theoretical physics. He wrote to his father to allow him to pursue theoretical physics. His father accepted his request on one condition. Bhabha would need to complete his mechanical engineering Tripos successfully. After his success Bhabha joined the famed Cavendish Laboratory as research scholar to pursue PhD in theoretical physics. During this period he travelled extensively in the European countries on travel studentships and met several great physicists and science luminaries of the times. Bhabha's interaction with leading physicists of the times reinforced his interest in the study of atoms, quantum theory and nuclear physics.

At Cambridge, Bhabha devoted much of the time in the study of cosmic rays. The collaboration of Bhabha and Walter Heitler, a German physicist, resulted in the celebrated 'cascade theory of electron showers'. The postulations by Bhabha-Heitler showed that when the components of cosmic rays enter the earth's atmosphere, they collide with the atoms in the earth's atmosphere. These collisions lead to a cascading effect and the number of particles increased after collisions. This led Bhabha to conclude that the very penetrating particles, observed at ground level and underground, should be mass and intermediate between the electrons and the protons. Bhabha investigated positron interactions using Dirac's hole theory. He explained the destruction of the fast positrons by electrons in the K shells. In collaboration with Hulme, Bhabha gave final touches to his theory of electron-positron scattering, a process which came to be known as 'Bhabha Scattering'.

In 1937 Bhabha predicted the 'meson', an unstable sub atomic particle, which is made up of quarks and anti quarks. Bhabha stated that positive (or negative) mesons should spontaneously decay into a positron (or electron). He proposed vector nature of the meson fields as against the original Yukawa's theory of mesons being scalar particles. He, in essence played a very critical and leading role in the early days of the development of quantum electrodynamics and paved the path for all scientists. In 1935, he received his PhD from Cambridge in theoretical physics.

In 1939, Bhabha came to India on a short holiday visit but could not return to Europe because of the pre World War-II situation. In India, he took up the position of Special Reader teaching theoretical physics at the I.I.Sc in Bangalore. He was committed to the growth of modern science in India. In 1941, he was elected a Fellow of the Royal Society, London. He came to realize the immense potential that science and technology had and the lasting role they alone could play to transform India to a developed country. He knew the value and importance of ground level research. With this view, Bhabha corresponded with J.R.D. Tata and Sir Sorab Saklatvala and sent a formal proposal to the Tata trust for support in establishing an institute promoting research in fundamental physics. His efforts resulted in the 'Tata Institute of Fundamental Research (TIFR) in 1945 with Bhabha as its director. It became not only a top of the class centre for basic scientific research but also acted as an incubator where new age industrial enterprises could be nourished.

Soon after the independence, a Board of Research on Atomic Energy was constituted as a part of Council for Scientific and Industrial Research (CSIR) with Bhabha as its chairman. Bhabha was successful in persuading Jawaharlal Nehru, the then Prime Minister to enact in Parliament the Atomic Energy Bill, creating the Atomic Energy Commission (AEC). Bhabha was made its first Chairman. In 1954 a separate Ministry of Atomic Energy was established and Bhabha was appointed Secretary to this Ministry. He played a very crucial role in drafting the Scientific Policy Resolution which was introduced in Parliament in 1958. Bhabha, with his open mind-set sent many scientists for training abroad to know the latest advancements going on in science. This helped the country immensely in terms of honing up the expertise and capabilities of the Indian scientists and kept them abreast with the latest developments around the world.

Bhabha was able to indigenously develop capabilities for the complete nuclear fuel cycle right from sourcing of Uranium etc. He was instrumental in installing nuclear reactors and their various components in India. He was also instrumental in defining the three stage program in 1950 - survey of natural resources, development of strong research scholars and starting of instrumentation units in electronics like ECIL in Hyderabad. He underlined the need for harnessing the amounts of nuclear and atomic energies only for peaceful purposes like power generation and medicine. He became Governor of the International Atomic Energy Agency at Vienna. He was also inducted as a member of scientific advisory committee to the secretary general of the UN.

Through out his life time, Bhabha achieved a lot in scientific research, discoveries, organizational constructions and nation building through the use of science. For his stellar contribution to the world of science, he was awarded honorary doctorates by many Indian and

foreign universities. In 1954 he got the prestigious Padma Bhushan award. In his tragic death, India lost one of the most valuable sons. But he had left behind an organization which could advance India in nuclear science and technology.

JAGADISH CHANDRABOSE

J.C.Bose occupies a unique position in the history of modern Indian science. He was born on 30 November 1858 in Mymensingh, now in Bangladesh. His family was quite affluent and his father Bhagavan Chandra Bose was highly educated and had a great appreciation for science as a subject. At the time when civil services were the most sought after occupation, he wanted his son to pursue science and become a scientist.

Bose's education started in a vernacular school because his father believed that one must have command over one's own mother tongue before beginning English and that one should know one's own culture, literature, people and traditions. Bose was sent to Calcutta (now Kolkata) to join the Hare school in 1869 and then St.Xavier's school. In 1875 after passing Entrance Examination he was admitted to St.Xavier's College. Subsequently he received a Bachelor's Degree from the University of Calcutta in the physical science group. He had to surmount many hindrances both financial and filial to go to England. In England Bose took up the study of medicine, however, was throughout afflicted by fever. Then he was advised to study science instead and he took up a course of natural science. He received the Natural Science Tripos from the University of Cambridge and also completed a B. Sc from the University of London in 1884.

When Bose returned to India from England, he ran from pillar to post to get a job. Finally, he was given the post of Officiating Professor of Physics at The Presidency College, Calcutta. He was allowed to draw half of the salary that English at the same position would get. He accepted the position but was amply perturbed by this discrimination and decided to put up fight against it. After three years of rigorous fight, Bose got his entire salary including arrears.

In 1884, he decided to pursue scientific research and not to be confined to teaching assignment alone. He had to conduct his researches in a small 24 square feet room which he was given in the Presidency College. He started making arrangements and with the help of an untrained tinsmith, he devised and constructed new apparatus for his first research on electric radiation. He devised and fabricated a new type of radiator for generating radio waves. He was able to invent a very sensitive "Coherer" which was a comprehensive improvement over those of both Branly and Lodge. With

this improved version of equipment Bose could show various properties of radio waves. He also demonstrated a new type of radio waves which are very small such as microwaves of today.

In May 1895 he presented his first research paper before the Asiatic Society of Bengal on the topic “The polarization of Electric Rays by Double Reflecting Crystals. Since then, he had been sending his research papers to many periodicals and societies. In January 1897, he was invited to deliver a lecture at the Royal Institute of London. In his lecture, Bose demonstrated his devices for generation and detection of radio waves. It was an instant hit and got praises from all the scientists present. It is a considered belief of many who were aware of the developments that Bose had created wireless communications as much as two years before Marconi who was formally credited to have discovered it and got Nobel Prize for it. Bose had to face several adverse situations both in professional and personal life. But he never succumbed to them.

The international adulation that Bose received for his success had a very deep impact in India. As an encouraging gesture a new post with higher emoluments, with more initiative and with reasonable leisure for research was created for Bose by the Lieutenant Governor of Bengal. He was sent on deputation to England for six months. At a meeting held at the British Association for Advancement of science at Liverpool, Bose discoursed on his new investigations and findings on radio waves. In 1897 he was again deputed to Great Britain for three months. When he was invited by the Royal Society of Science to deliver a lecture on “On the Polarization of Electric Rays”. The lecture was widely appreciated for its content. Bose became an instant hit in the scientific circles of Europe including France and Germany.

Bose’s peers in England were extremely impressed by the achievements and helped Bose to set up a laboratory. Bose was against the philosophy of patenting and he wanted to gain no personal advantage from his inventions. In 1901 Bose refused to sign a commercial agreement for production of his apparatus.

Bose was pretty fascinated by some of the peculiar responses that his electric wave receiver exhibited. This led Bose to assume that metals too had “feelings” and sensitivity. Bose observed that plants responded more favourably to his experiments and established the unity between living and non-living. In 1900, he was invited to the Paris International Congress of Physicists where he read his paper on “The Similarity Responses of Inorganic and Living Matter.” His paper was considered as one of the most path breaking discoveries of the phenomena. Swamy Vivekananda, who was in Paris at that time, went to listen Bose’s discourse and felt very proud. In 1901 Bose read a paper at the Royal Society on the responses of living and non-living to stimuli. Though his

ideas were opposed by some plant physiologists, Bose decided to prolong his stay in London to conduct further experiments and prove his theory. Eventually, Bose returned to Kolkata and continued his research by inventing very sensitive instruments to help him in his research. One such instrument was Crescograph, an instrument meant to measure the growth of plant in a very microscopic way. His findings had wide implications and eventually influenced many subjects.

In 1903, Bose was honoured with the coveted honorary title of the commander of Order of the Indian Empire (CIE) and in 1912 he was awarded the Commander of the Star of India (CSI). He was also given the title of “Sir.” On 23 November 1937, he left for heavenly abode leaving behind a legacy which will keep the country’s head high for years to come.

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